STRUCTURE AND EXPLICIT K-THEORY OF THE BUNCE-DEDDENS SHIFT ALGEBRA

SHELLEY HEBERT, SLAWOMIR KLIMEK, MATT MCBRIDE, J. WILSON PEOPLES

Allowed standard prerequisites (editorial).

- Research prerequisites named in the original article. Full original bibliography retained; all source-local definitions and supporting arguments retained.

1. Introduction

Let $\mathbb{Z}_s$ be the ring of s-adic integers, where s is a positive integer, not necessarily a prime. Let $\mathcal{V}$ be the set of balls in $\mathbb{Z}_s$. The corresponding Hilbert space $\ell^2(\mathcal{V})$ carries a natural, diagonal representation of the algebra of continuous functions on $\mathbb{Z}_s$. The purpose of this paper and the companion paper [12] is to study natural C^*-algebras in $\ell^2(\mathcal{V})$ generated by $C(\mathbb{Z}_s)$ along with specific shifts. Those shift algebras are interesting examples of C^*-algebras with connections to number theory and combinatorics, a subject that has gained some popularity, see for example [9, 25].

Our motivation to study C^*-algebras associated with shifts on trees partially comes from trying to extend the results from [17, 19, 20] on p-adic spectral triples as well as ideas from [23] on noncommutative geometry of ultrametric Cantor sets. A heuristic goal is to understand "quantum" non-archimedean spaces and their geometry with seemingly many candidates for such concepts.

We review all the definitions and the key results from the companion paper [12] so that the present paper is self-contained. The main purpose of [12] was to determine how each of those shift algebras can be identified with a crossed product by an endomorphism, in fact a monomorphism with a hereditary range. In this paper, we delve deeper into the structure of the shift algebras and compute their K-theory.

There are several competing concepts of crossed product by an endomorphism [1, 3, 11, 22, 29] but they all agree for monomorphisms with a hereditary range. In [12], we took a pragmatic approach to crossed products by finding enlargements of the initial algebra $C(\mathbb{Z}_s)$ up to a better behaved "coefficient" C^*-subalgebras of the shift algebras which makes the corresponding monomorphisms to have hereditary range. Those coefficient algebras are the starting point of the current paper.

To understand the structure of the shift algebras, we consider natural ideals in each of the algebras and then describe their quotients. The main technique is an analog of a basic construction for usual crossed products by automorphisms involving invariant ideals, see [30, Section 3.4]. In particular, we prove a counterpart to [30, Proposition 3.19] which gives a short exact sequence of crossed products obtained from an invariant ideal.

Another key technique used in this paper to understand the structure of some crossed products is to make them concrete algebras through faithful representations. Faithfulness is established using a version of the O'Donovan condition easily implied by the existence of gauge actions.

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A short exact sequence of shift C^*-algebras induces a 6-term exact sequence in K-theory which we then, in each case, analyze and reduce to short exact sequences. This is enough to determine the K-groups in all but the Bunce–Deddens shift case for which the sequence does not split. Instead, in this case, we describe the K-theory as a direct limit of finitely generated free Abelian groups.

The paper is organized as follows. Preliminary Section 2 contains the definitions of shift algebras and a gauge action for each. Crossed products by endomorphisms, coefficient algebras with gauge actions and invariant ideals are the subject of Section 3. Each of the following four sections is devoted to one of the shift algebras. Those sections are structured similarly: we recall the coefficients algebras from [12], describe the coefficient algebra ideals, their quotients and the quotient endomorphisms, followed by the identification of the corresponding shift algebra ideals and their quotients. Each section is then concluded with a calculation of the K-theory.

2. Preliminaries

2.1. **s-adic integers.** For $s \in \mathbb{N}$ with $s \geq 2$, the space of s-adic integers $\mathbb{Z}_s$ is defined as consisting of infinite sums

$$\mathbb{Z}_s = \left\{ x = \sum_{j=0}^{\infty} x_j s^j \mid 0 \leq x_j \leq s-1,\ j = 0, 1, 2, \dots \right\}.$$

We call the above expansion of x its s-adic expansion. Note that $\mathbb{Z}_s$ is isomorphic with the countable product $\prod \mathbb{Z}/s\mathbb{Z}$ and so, when the product is equipped with the Tychonoff topology, it is a Cantor set. It is also a metric space with the usual norm $|\cdot|_s$, which is defined as follows $|0|_s = 0$ and, if $x \neq 0$, $|x|_s = s^{-n}$, where x_n is the first nonzero term in the above s-adic expansion. Moreover, $\mathbb{Z}_s$ is an Abelian ring with unity with respect to addition and multiplication with a carry and we have $\mathbb{Z}_s \cong \varprojlim \mathbb{Z}/s^n\mathbb{Z}$. Using the division algorithm modulo s repeatedly, we can associate an s-adic expansion to any integer and so $\mathbb{Z}_s$ naturally contains a copy of $\mathbb{Z}$ which is dense in $\mathbb{Z}_s$.

Any ball in $\mathbb{Z}_s$ is of the form

$$\left\{ y \in \mathbb{Z}_s \mid |y - x|_s \leq \frac{1}{s^n} \right\},$$

where $n \in \mathbb{Z}_{\geq 0}$ and $0 \leq x < s^n$. Here x is a particular center of this ball. Consequently, we can write the set of balls $\mathcal{V}$ in the following way $\mathcal{V} = \{(n, x) \mid n = 0, 1, 2, \dots,\ 0 \leq x < s^n\}$.

2.2. **Hilbert spaces and shifts.** Throughout this paper, we use the Hilbert space $H = \ell^2(\mathcal{V})$ with canonical basis $\{E_{(n,x)}\}$, where $n = 0, 1, 2, \dots$ and $0 \leq x < s^n$. Let $f \in C(\mathbb{Z}_s)$ and define a bounded operator $M_f \colon H \to H$ by $M_f E_{(n,x)} = f(x) E_{(n,x)}$. Due to the density of $\mathbb{Z}_{\geq 0}$ in $\mathbb{Z}_s$, we have that

$$\|M_f\| = \sup_{x \in \mathbb{Z}_{\geq 0}} |f(x)| = \sup_{x \in \mathbb{Z}_s} |f(x)|$$

and so $f \mapsto M_f$ is a faithful representation of the C^*-algebra $C(\mathbb{Z}_s)$.

There are four main shifts in H that are the focus of this paper and which we call the Bunce–Deddens shift, the Hensel shift, the Bernoulli shift, and the Serre shift. All four shifts are defined on H with formulas given on the basis elements of H as follows:

(1) *The Bunce–Deddens shift:* $U \colon H \to H$ via $U E_{(n,x)} = E_{(n+1,x+1)}$.
(2) *The Hensel shift:* $V \colon H \to H$ via $V E_{(n,x)} = E_{(n+1,sx)}$.

(3) *The Bernoulli shift:* $S\colon H \to H$ via

$$SE_{(n,x)} = \frac{1}{s}\sum_{j=0}^{s-1} E_{(n+1,sx+j)}.$$

(4) *The Serre shift:* $W\colon H \to H$ via

$$WE_{(n,x)} = \frac{1}{\sqrt{s}}\sum_{j=0}^{s-1} E_{(n+1,x+js^n)}.$$

A computation yields their respective adjoints

$$(1)\quad U^*E_{(n,x)} = \begin{cases} E_{(n-1,x-1)} & \text{if } n \geq 1,\ 0 < x \leq s^{n-1}, \\ 0 & \text{if } n = 0 \text{ or } x = 0 \text{ or } s^{n-1} < x < s^n, \end{cases}$$

$$(2)\quad V^*E_{(n,x)} = \begin{cases} E_{(n-1,x/s)} & \text{if } n \geq 1,\ s \mid x, \\ 0 & \text{if } n = 0 \text{ or } s \nmid x, \end{cases}$$

$$(3)\quad S^*E_{(n,x)} = \begin{cases} E_{(n-1,(x-x \bmod s)/s)} & \text{if } n \geq 1, \\ 0 & \text{if } n = 0, \end{cases}$$

$$(4)\quad W^*E_{(n,x)} = \begin{cases} \dfrac{1}{\sqrt{s}} E_{(n-1,x \bmod s^{n-1})} & \text{if } n \geq 1, \\ 0 & \text{if } n = 0. \end{cases}$$

It is easy to verify that $U^*U = V^*V = S^*S = W^*W = I$, hence each one is an isometry. For each of these four shifts $\mathcal{J} \in \{U, V, S, W\}$, we have the corresponding shift C^*-algebras

$$A_{\mathcal{J}} := C^*(\mathcal{J}, M_f \mid f \in C(\mathbb{Z}_s)),$$

which are the main objects of interest in this paper.

By analogy with graph algebras, a gauge action is an important structure shared by all those shift algebras. For $\theta \in \mathbb{R}/\mathbb{Z}$, consider the following strongly continuous one-parameter family $\{\mathcal{U}_\theta\}$ of unitary operators on $\ell^2(\mathcal{V})$ defined on basis elements by $\mathcal{U}_\theta E_{(n,x)} = \mathrm{e}^{2\pi i n\theta} E_{(n,x)}$. We define a continuous one-parameter family $\{\rho_\theta\}$ of automorphisms of $A_{\mathcal{J}}$, the gauge action $\rho_\theta\colon A_{\mathcal{J}} \to A_{\mathcal{J}}$, by the formula $\rho_\theta(a) = \mathcal{U}_\theta a \mathcal{U}_\theta^{-1}$. Then on generators of $A_{\mathcal{J}}$ we have

$$\rho_\theta(\mathcal{J}) = \mathrm{e}^{2\pi i\theta}\mathcal{J}, \qquad \rho_\theta(\mathcal{J}^*) = \mathrm{e}^{-2\pi i\theta}\mathcal{J}^*, \qquad \text{and} \qquad \rho_\theta(M_f) = M_f.$$

Throughout the rest of this paper, we describe the structure of each algebra and study its K-theory.

3. Coefficient algebras with a gauge action

3.1. **Coefficient algebras.** The notion of a coefficient algebra was introduced in [21]. A review of this concept, with additional material suitable for our purposes, is the main theme of this section.

Given a Hilbert space $\mathcal{H}$, let $\mathcal{J} \in B(\mathcal{H})$ be an isometry and let $B_{\mathcal{J}} \subseteq B(\mathcal{H})$ be a C^*-subalgebra of $B(\mathcal{H})$ containing the identity I. We use the notation $A_{\mathcal{J}}$ for the C^*-algebra generated by $B_{\mathcal{J}}$ and $\mathcal{J}$. We call $B_{\mathcal{J}}$ a *coefficient algebra* of the C^*-algebra $A_{\mathcal{J}}$ if $B_{\mathcal{J}}$ and $\mathcal{J}$ satisfy the following conditions $\alpha_{\mathcal{J}}(a) := \mathcal{J}a\mathcal{J}^* \in B_{\mathcal{J}}$ and $\beta_{\mathcal{J}}(a) := \mathcal{J}^*a\mathcal{J} \in B_{\mathcal{J}}$, for all $a \in B_{\mathcal{J}}$.

It is easy to see that $\alpha_{\mathcal{J}}$ is an endomorphism of $B_{\mathcal{J}}$ and we have the following properties $\beta_{\mathcal{J}}\alpha_{\mathcal{J}}(a) = a$, $\alpha_{\mathcal{J}}\beta_{\mathcal{J}}(a) = \alpha_{\mathcal{J}}(I)a\alpha_{\mathcal{J}}(I)$, and $\beta_{\mathcal{J}}(I) = I$. Moreover, the map $\beta_{\mathcal{J}}\colon B_{\mathcal{J}} \to B_{\mathcal{J}}$ is linear, continuous, positive and has the "transfer" property $\beta_{\mathcal{J}}(\alpha_{\mathcal{J}}(a)b) = a\beta_{\mathcal{J}}(b)$, and thus is an example of Exel's transfer operator [11]. Also notice that if the algebra $B_{\mathcal{J}}$ is commutative, or more generally if $\alpha_{\mathcal{J}}(I)$

is in the center of $B_{\mathcal{J}}$, then $\beta_{\mathcal{J}}$ is also an endomorphism. Indeed, notice that $\alpha_{\mathcal{J}}(I) = \mathcal{J}\mathcal{J}^*$ and we have that

$$\beta_{\mathcal{J}}(a)\beta_{\mathcal{J}}(b) = \mathcal{J}^* a \mathcal{J}\mathcal{J}^* b\mathcal{J} = \mathcal{J}^* a \alpha_{\mathcal{J}}(I) b\mathcal{J} = \mathcal{J}^* \alpha_{\mathcal{J}}(I) ab\mathcal{J} = \mathcal{J}^*\mathcal{J}\mathcal{J}^* ab\mathcal{J} = \beta_{\mathcal{J}}(ab).$$

Clearly, $\alpha_{\mathcal{J}}(I)$ is a projection and notice that $\mathrm{Ran}(\alpha_{\mathcal{J}})$, the range of $\alpha_{\mathcal{J}}$, is a hereditary subalgebra of $B_{\mathcal{J}}$ and we have $\mathrm{Ran}(\alpha_{\mathcal{J}}) = \alpha_{\mathcal{J}}(I) B_{\mathcal{J}} \alpha_{\mathcal{J}}(I)$.

3.2. **Gauge action.** We now consider additional structure called a gauge action. Assume that there is a one-parameter family $\{\mathcal{U}_\theta\}$, for $\theta \in \mathbb{R}/\mathbb{Z}$, of unitary operators on $\mathcal{H}$ satisfying the following commutation properties $\mathcal{U}_\theta a = a\mathcal{U}_\theta$, $\mathcal{U}_\theta \mathcal{J} = \mathrm{e}^{2\pi \mathrm{i}\theta}\mathcal{J}\mathcal{U}_\theta$ for all $a \in B_{\mathcal{J}}$. We define a one-parameter family $\{\rho_\theta\}$ of automorphisms $\rho_\theta \colon B(\mathcal{H}) \to B(\mathcal{H})$ by the formula $\rho_\theta(a) = \mathcal{U}_\theta a \mathcal{U}_\theta^{-1}$. It follows that on generators of $A_{\mathcal{J}}$ we have

$$\rho_\theta(\mathcal{J}) = \mathrm{e}^{2\pi\mathrm{i}\theta}\mathcal{J}, \qquad \rho_\theta(\mathcal{J}^*) = \mathrm{e}^{-2\pi\mathrm{i}\theta}\mathcal{J}^*, \qquad \text{and} \qquad \rho_\theta(a) = a, \quad a \in B_{\mathcal{J}}.$$

Clearly, the map $\theta \mapsto \rho_\theta(x)$ is continuous for all x which are polynomials in $\mathcal{J}$ and $\mathcal{J}^*$ with coefficients in $B_{\mathcal{J}}$. It follows by an approximation argument that it is continuous for every $x \in A_{\mathcal{J}}$.

As a consequence of the existence of a gauge action, we have an expectation $E \colon A_{\mathcal{J}} \to A_{\mathcal{J}}$ given by the formula

$$E(x) := \int_0^1 \rho_\theta(x) \mathrm{d}\theta. \tag{3.1}$$

The authors of [12] noted that $C(\mathbb{Z}_s)$ is not a coefficient algebra for the shift algebras currently considered. Instead, coefficient algebra extensions of $C(\mathbb{Z}_s)$ were constructed as the invariant subalgebras $A_{\mathcal{J},\mathrm{inv}}$ of $A_{\mathcal{J}}$ with respect to the automorphism ρ_θ. This is because $C(\mathbb{Z}_s) \subseteq A_{\mathcal{J},\mathrm{inv}}$ and, for every $a \in A_{\mathcal{J},\mathrm{inv}}$, we also have $\mathcal{J}^* a \mathcal{J} \in A_{\mathcal{J},\mathrm{inv}}$ and $\mathcal{J} a \mathcal{J}^* \in A_{\mathcal{J},\mathrm{inv}}$.

3.3. **O'Donovan condition.** Using the commutation relation $\mathcal{J}a = \alpha_{\mathcal{J}}(a)\mathcal{J}$, one can arrange the polynomials in $\mathcal{J}$, $\mathcal{J}^*$ with coefficients in $B_{\mathcal{J}}$ so that the coefficients are in front of the powers of $\mathcal{J}$ and behind the powers of $\mathcal{J}^*$. This leads to the following observation (see [21, Proposition 2.3]): the vector space $\mathcal{F}(B_{\mathcal{J}}, \mathcal{J})$ consisting of finite sums

$$x = \sum_{n \geq 0} a_n \mathcal{J}^n + \sum_{n<0} (\mathcal{J}^*)^{-n} a_n, \tag{3.2}$$

where $a_n \in B_{\mathcal{J}}$, is a dense $*$-subalgebra of $A_{\mathcal{J}}$. Additionally, using $\mathcal{J}^n = \mathcal{J}^n(\mathcal{J}^*)^n\mathcal{J}^n$ and its adjoint one can always choose coefficients a_n such that $a_n = a_n\mathcal{J}^n(\mathcal{J}^*)^n$, $n \geq 0$ and $a_n = \mathcal{J}^{-n}(\mathcal{J}^*)^{-n}a_n$, $n < 0$.

Then, applying the expectation defined in equation (3.1), we obtain

$$a_n = E(x(\mathcal{J}^*)^n), \quad n \geq 0 \qquad \text{and} \qquad a_n = E(\mathcal{J}^{-n}x), \quad n < 0, \tag{3.3}$$

and thus such coefficients a_n are unique. More generally, by the above formulas we can associate "Fourier" coefficients $a_n \in B_{\mathcal{J}}$ to any $x \in A_{\mathcal{J}}$ and, by the usual Cesaro summation argument [15], the coefficients a_n fully determine x.

We also have $\|a_0\| = \|E(x)\| \leq \|x\|$, which is called the O'Donovan condition. A key result (see [21, Theorem 2.7]) says that since $A_{\mathcal{J}}$ satisfies the O'Donovan condition then $A_{\mathcal{J}}$ is isomorphic to the crossed product C^*-algebra $B_{\mathcal{J}} \rtimes_{\alpha_{\mathcal{J}}} \mathbb{N}$

$$A_{\mathcal{J}} \cong B_{\mathcal{J}} \rtimes_{\alpha_{\mathcal{J}}} \mathbb{N}, \tag{3.4}$$

see [3] and the appendix of [13]. For concreteness, we use Stacey's "multiplicity-one crossed product", see [3]. Namely, $B_{\mathcal{J}} \rtimes_{\alpha_{\mathcal{J}}} \mathbb{N}$ is defined to be the universal C^*-algebra generated by a copy of $B_{\mathcal{J}}$ and an isometry $\mathcal{L}$ such that for all $a \in B_{\mathcal{J}}$, $\mathcal{L}a\mathcal{L}^* = \alpha_{\mathcal{J}}(a)$.

The significance of the equation (3.4) is that the crossed product on the right-hand side is a universal object, defined in terms of generators and relations, and thus it gives us a way to construct convenient, concrete representations. A representation of the crossed product $B_{\mathcal{J}} \rtimes_{\alpha_{\mathcal{J}}} \mathbb{N}$ is faithful if it is faithful on $B_{\mathcal{J}}$ and satisfies the O'Donovan condition [3]. In particular, such a representation is faithful if it is faithful on $B_{\mathcal{J}}$ and admits a gauge action.

3.4. **Invariant ideals.** In this section, we consider invariant ideals in coefficient algebras and the structure they impose on the crossed products, analogously to the usual crossed products by automorphisms, see [30, Section 3.4]. While this is standard material, for completeness we provide short proofs.

Let $I_{\mathcal{J}}$ be an $\alpha_{\mathcal{J}}$ and $\beta_{\mathcal{J}}$ invariant ideal in $B_{\mathcal{J}}$ and let $\mathcal{I}_{\mathcal{J}}$ be the subalgebra in $A_{\mathcal{J}}$ defined as $\mathcal{I}_{\mathcal{J}} := C^*(a\mathcal{J}^n \mid a \in I_{\mathcal{J}}, n = 0, 1, \dots)$. Note that $I_{\mathcal{J}} \subseteq \mathcal{I}_{\mathcal{J}}$. Alternatively, one can define $\mathcal{I}_{\mathcal{J}}$ by using the Fourier coefficients a_n from equations (3.2) and (3.3) $x \in \mathcal{I}_{\mathcal{J}}$ if and only if for all $n \in \mathbb{Z}$: $a_n \in I_{\mathcal{J}}$.

We have the following elementary fact.

Proposition 3.1. *The algebra $\mathcal{I}_{\mathcal{J}}$ is an ideal in $A_{\mathcal{J}}$.*

Proof. We write the generators more explicitly $A_{\mathcal{J}} = C^*(b, \mathcal{J}, \mathcal{J}^* \mid b \in B_{\mathcal{J}})$ and similarly $\mathcal{I}_{\mathcal{J}} = C^*(a\mathcal{J}^n, (\mathcal{J}^*)^n a \mid a \in I_{\mathcal{J}}, n = 0, 1, \dots)$. It is sufficient to verify that the products of generators are in $\mathcal{I}_{\mathcal{J}}$. This requires checking multiple cases.

Let $a \in I_{\mathcal{J}}$, $b \in B_{\mathcal{J}}$. It is clear that $ba\mathcal{J}^n, a\mathcal{J}^n\mathcal{J}$, and $\mathcal{J}a\mathcal{J}^n$ are in $\mathcal{I}_{\mathcal{J}}$. Next, note that $a\mathcal{J}^n b = a\alpha_{\mathcal{J}}^n(b)\mathcal{J}^n \in \mathcal{I}_{\mathcal{J}}$ since $a\alpha_{\mathcal{J}}^n(b) \in I_{\mathcal{J}}$.

For adjoints, we have

$$\mathcal{J}^* a \mathcal{J}^n = \begin{cases} \mathcal{J}^* a \mathcal{J} \mathcal{J}^{n-1}, & n > 0, \\ \mathcal{J}^* a, & n = 0 \end{cases} = \begin{cases} \beta_{\mathcal{J}}(a)\mathcal{J}^{n-1}, & n > 0, \\ \mathcal{J}^*, a & n = 0 \end{cases} \in \mathcal{I}_{\mathcal{J}}$$

since $\beta_{\mathcal{J}}(a) \in I_{\mathcal{J}}$. Further, note that $\mathcal{J}\mathcal{J}^* = \alpha_{\mathcal{J}}(1) \in B_{\mathcal{J}}$ so

$$a\mathcal{J}^n \mathcal{J}^* = \begin{cases} a\mathcal{J}^{n-1}\alpha_{\mathcal{J}}(1), & n > 0, \\ a\mathcal{J}^*, & n = 0 \end{cases} = \begin{cases} a\alpha_{\mathcal{J}}^n(1)\mathcal{J}^{n-1}, & n > 0, \\ \mathcal{J}^*\alpha_{\mathcal{J}}(a), & n = 0 \end{cases} \in \mathcal{I}_{\mathcal{J}}.$$

Finally, $(\mathcal{J}^*)^n ab \in \mathcal{I}_{\mathcal{J}}$, $b(\mathcal{J}^*)^n a = (\mathcal{J}^*)^n \alpha_{\mathcal{J}}^n(b) a \in \mathcal{I}_{\mathcal{J}}$, so $\mathcal{I}_{\mathcal{J}}$ is an ideal completing the proof. □

Next consider the factor maps $[\alpha_{\mathcal{J}}], [\beta_{\mathcal{J}}] \colon B_{\mathcal{J}}/I_{\mathcal{J}} \to B_{\mathcal{J}}/I_{\mathcal{J}}$ defined by

$$[\alpha_{\mathcal{J}}](a + I_{\mathcal{J}}) = \alpha_{\mathcal{J}}(a) + I_{\mathcal{J}} \qquad \text{and} \qquad [\beta_{\mathcal{J}}](a + I_{\mathcal{J}}) = \beta_{\mathcal{J}}(a) + I_{\mathcal{J}}.$$

Proposition 3.2. *$[\alpha_{\mathcal{J}}]$ is a monomorphism of $B_{\mathcal{J}}/I_{\mathcal{J}}$ with hereditary range.*

Proof. Note that $\alpha_{\mathcal{J}}$ is an endomorphism of $B_{\mathcal{J}}$, thus $[\alpha_{\mathcal{J}}]$ is an endomorphism of $B_{\mathcal{J}}/I_{\mathcal{J}}$. Further, $\beta_{\mathcal{J}} \circ \alpha_{\mathcal{J}} = I$, so $[\beta_{\mathcal{J}}] \circ [\alpha_{\mathcal{J}}] = [I]$, thus $[\alpha_{\mathcal{J}}]$ is a monomorphism.

For the range of $[\alpha_{\mathcal{J}}]$, we know from [12] that $\mathrm{Ran}\alpha_{\mathcal{J}} = \alpha_{\mathcal{J}}(I)B_{\mathcal{J}}\alpha_{\mathcal{J}}(I)$. This implies that $\mathrm{Ran}[\alpha_{\mathcal{J}}] = [\alpha_{\mathcal{J}}][I]B_{\mathcal{J}}/I_{\mathcal{J}}[\alpha_{\mathcal{J}}][I]$ and so $[\alpha_{\mathcal{J}}]$ has hereditary range. □

We now investigate the structure of the factor algebra $A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}$.

Proposition 3.3. *We have the following isomorphism of algebras:*

$$B_{\mathcal{J}}/I_{\mathcal{J}} \rtimes_{[\alpha_{\mathcal{J}}]} \mathbb{N} \cong A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}.$$

Proof. $B_{\mathcal{J}} \rtimes_{[\alpha_{\mathcal{J}}]} \mathbb{N}$ is the universal C^*-algebra generated by a copy of $B_{\mathcal{J}}/I_{\mathcal{J}}$ and an isometry $\mathcal{L}_{\mathcal{J}}$ such that $\mathcal{L}_{\mathcal{J}}[a]\mathcal{L}_{\mathcal{J}}^* = [\alpha_{\mathcal{J}}][a]$.

Define the map $\varphi_{\mathcal{J}} \colon B_{\mathcal{J}}/I_{\mathcal{J}} \rtimes_{[\alpha_{\mathcal{J}}]} \mathbb{N} \to A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}$ on generators by

$$\varphi_{\mathcal{J}}(a + I_{\mathcal{J}}) = a + \mathcal{I}_{\mathcal{J}} \qquad \text{and} \qquad \varphi_{\mathcal{J}}(\mathcal{L}_{\mathcal{J}}) = \mathcal{J} + \mathcal{I}_{\mathcal{J}}.$$

This is well-defined as $I_{\mathcal{J}} \subseteq \mathcal{I}_{\mathcal{J}}$. Since $\mathcal{J}$ is an isometry, so is $\varphi_{\mathcal{J}}(\mathcal{L}_{\mathcal{J}})$. Furthermore, the map $\varphi_{\mathcal{J}}$ preserves the crossed product relation

$$\varphi_{\mathcal{J}}(\mathcal{L}_{\mathcal{J}})\varphi_{\mathcal{J}}(a+I_{\mathcal{J}})\varphi_{\mathcal{J}}(\mathcal{L}_{\mathcal{J}})^* = \mathcal{J}a\mathcal{J}^* + \mathcal{I}_{\mathcal{J}} = \alpha_{\mathcal{J}}(a) + \mathcal{I}_{\mathcal{J}} = [\alpha_{\mathcal{J}}]\varphi_{\mathcal{J}}(a+I_{\mathcal{J}}).$$

By universality, $\varphi_{\mathcal{J}}$ extends to a homomorphism from $B_{\mathcal{J}}/I_{\mathcal{J}} \rtimes_{[\alpha_{\mathcal{J}}]} \mathbb{N}$ to $A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}$.

By uniqueness of the Fourier coefficients of a, see equation (3.3), we see that if $a \in \mathcal{I}_{\mathcal{J}}$ with $a \in B_{\mathcal{J}}$, then $a \in I_{\mathcal{J}}$. This means that the map $\varphi_{\mathcal{J}}$ is faithful on $B_{\mathcal{J}}/I_{\mathcal{J}}$. To verify the map is faithful on the whole crossed product, we observe that the algebra $A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}$ admits a gauge action. Indeed, since ρ_θ preserves $I_{\mathcal{J}} \subseteq B_{\mathcal{J}}$ it factors to the quotient $[\rho]_\theta \colon A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}} \to A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}$. We have that $[\rho]_\theta$ is a 1-parameter family of automorphisms with

$$[\rho]_\theta(a+\mathcal{I}_{\mathcal{J}}) = a+\mathcal{I}_{\mathcal{J}} \quad \text{for } a \in B_{\mathcal{J}}, \qquad [\rho]_\theta(\mathcal{J}+\mathcal{I}_{\mathcal{J}}) = \mathrm{e}^{2\pi \mathrm{i}\theta}\mathcal{J}+\mathcal{I}_{\mathcal{J}}$$

and therefore $\varphi_{\mathcal{J}}$ is faithful by [3].

Finally, notice that $\varphi_{\mathcal{J}}$ is onto because its range is closed and it contains the generators of $A_{\mathcal{J}}/\mathcal{I}_{\mathcal{J}}$. □

As a consequence, we get the following short exact sequence of C^*-algebras, which is the basis for all structural results in the following sections:

$$0 \to \mathcal{I}_{\mathcal{J}} \to A_{\mathcal{J}} \to B_{\mathcal{J}}/I_{\mathcal{J}} \rtimes_{[\alpha_{\mathcal{J}}]} \mathbb{N} \to 0. \tag{3.5}$$

4. The Bunce–Deddens shift

4.1. **The algebra A_U.** In this section, we consider the first shift algebra $A_U = C^*(U, M_f \mid f \in C(\mathbb{Z}_s))$, where the shift U is given on the basis elements of $H = \ell^2(\mathcal{V})$ by $UE_{(n,x)} = E_{(n+1,x+1)}$. The corresponding maps on A_U are $\alpha_U(a) := UaU^* \in A_U$ and $\beta_U(a) := U^*aU \in A_U$, for $a \in A_U$.

We define the maps $\mathfrak{a}_U$ and $\mathfrak{b}_U$ on $C(\mathbb{Z}_s)$ as follows: $\mathfrak{a}_U f(x) = f(x-1)$ and $\mathfrak{b}_U f(x) = f(x+1)$. Both are automorphisms on $C(\mathbb{Z}_s)$ and we have $\mathfrak{a}_U(1) = \mathfrak{b}_U(1) = 1$ and $\mathfrak{a}_U \circ \mathfrak{b}_U = \mathfrak{b}_U \circ \mathfrak{a}_U = 1$. The operators U and M_f satisfy the following relations:

$$M_f U = U M_{\mathfrak{b}_U f} \qquad \text{and} \qquad \alpha_U(M_f) = M_{\mathfrak{a}_U f} U U^*.$$

It follows additionally that $UM_f = M_{\mathfrak{a}_U f}U$ and $\beta_U(M_f) = M_{\mathfrak{b}_U f}$.

In parallel, we will also describe and analyze the subalgebras

$$A_{U,n} := C^*(U, M_{(n,x)} \mid 0 \le x < s^n),$$

where $M_{(n,x)}$ are the multiplication operators by the characteristic functions of balls

$$M_{(n,x)}E_{(m,y)} = \begin{cases} E_{(m,y)} & \text{if } |y-x|_s \le s^{-n}, \\ 0 & \text{otherwise.} \end{cases}$$

We have the following formulas:

$$\mathfrak{a}_U M_{(n,x)} = M_{(n,x+1)} \qquad \text{and} \qquad \mathfrak{b}_U M_{(n,x)} = M_{(n,x-1)}.$$

Notice that $A_{U,n}$ is a subalgebra of $A_{U,n+1}$ and A_U is the C^*-algebras direct limit of $A_{U,n}$ via the inclusion maps $A_U = \varinjlim A_{U,n}$.

4.2. **The coefficient algebra of A_U.** Define the mutually orthogonal projections $\{P_n\}_{n\geq 0}$ by $P_0 = I - UU^*$ and $P_n = U^n P_0 (U^*)^n$. By [12, Proposition 4.2], the C^*-algebra

$$B_U := C^*(M_f, P_n \mid f \in C(\mathbb{Z}_s),\, n \in \mathbb{Z}_{\geq 0})$$

is the smallest coefficient algebra of A_U containing $C(\mathbb{Z}_s)$. The algebra B_U is commutative and by [12, Proposition 4.3] every element of B_U can be uniquely written as

$$T_U(F) := \sum_{n=0}^{\infty} (M_{f_n} - M_{f_\infty}) P_n + M_{f_\infty},$$

where $F = (f_n)_{n\in\mathbb{Z}_{\geq 0}} \in c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$, the space of uniformly convergent sequences of continuous functions on $\mathbb{Z}_s$, and

$$C(\mathbb{Z}_s) \ni f_\infty := \lim_{n\to\infty} f_n. \tag{4.1}$$

We will refer to such a T_U as a Toeplitz map.

Consequently, B_U is isomorphic to $c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$.

Put $B_{U,n} := C^*(P_k, M_{(n,x)} \mid k = 0, 1, \ldots, 0 \leq x < s^n)$. This is an Abelian coefficient algebra for $A_{U,n}$. We can identify $B_{U,n} \cong c(\mathbb{Z}_{\geq 0}, \mathbb{C}^{s^n})$, since $C^*(M_{(n,x)} \mid 0 \leq x < s^n) \cong \mathbb{C}^{s^n}$.

Consider the algebra I_U generated by the products

$$I_U := C^*(M_f P_n \mid f \in C(\mathbb{Z}_s),\, n \in \mathbb{Z}_{\geq 0}).$$

Clearly, I_U is an ideal in B_U and, using the above Toeplitz map T_U, we see that we have an identification $I_U \cong c_0(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$, where $c_0(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$ is consisting of those sequences $F = (f_n)_{n\in\mathbb{Z}_{\geq 0}}$ for which $f_\infty = 0$.

Similarly, define

$$I_{U,n} := C^*(P_k M_{(n,x)} \mid k \in \mathbb{Z}_{\geq 0},\, 0 \leq x < s^n) \cong c_0\big(\mathbb{Z}_{\geq 0}, \mathbb{C}^{s^n}\big).$$

Then $I_{U,n}$ is an ideal in $B_{U,n}$.

Consider the morphisms $\tilde{\alpha}_U$, $\tilde{\beta}_U$ on $c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$ given by

$$(\tilde{\alpha}_U F)_n(x) = \begin{cases} f_{n-1}(x-1) & \text{if } n \geq 0, \\ 0 & \text{if } n = 0 \end{cases} \qquad \text{and} \qquad \big(\tilde{\beta}_U F\big)_n(x) = f_{n+1}(x+1).$$

A simple calculation shows that we have the following relations:

$$\alpha_U(T_U(F)) = T_U\big(\tilde{\beta}_U F\big) \qquad \text{and} \qquad \beta_U(T_U(F)) = T_U(\tilde{\alpha}_U F).$$

It follows that the ideals I_U and $I_{U,n}$ are α_U and β_U invariant.

We have the following short exact sequence

$$0 \to c_0(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \to c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \to C(\mathbb{Z}_s) \to 0,$$

where the quotient map is $F \mapsto f_\infty \in C(\mathbb{Z}_s)$. It follows that the factor algebra B_U/I_U is isomorphic to $C(\mathbb{Z}_s)$ $B_U/I_U \cong C(\mathbb{Z}_s)$.

The morphisms $\tilde{\alpha}_U$, $\tilde{\beta}_U$ preserve the ideal $c_0(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$ and so they factor to morphisms on $C(\mathbb{Z}_s)$ which are exactly the automorphisms $\mathfrak{a}_U$ and $\mathfrak{b}_U$. We also get

$$B_{U,n}/I_{U,n} \cong C^*(M_{(n,x)}) \cong \mathbb{C}^{s^n}.$$

4.3. **The Structure of A_U.** The Bunce–Deddens algebra $BD(s^\infty)$ corresponding to the supernatural number s^∞, see [4, 5, 16], is the following crossed product by an automorphism $BD(s^\infty) = C(\mathbb{Z}_s) \rtimes_{\mathfrak{a}_U} \mathbb{Z}$. The goal is to relate A_U with the above Bunce–Deddens algebra.

From [12, Theorem 4.4], we have an isomorphism of C^*-algebras $A_U \cong c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \rtimes_{\tilde{\alpha}_U} \mathbb{N}$. We also have $A_{U,n} \cong B_{U,n} \rtimes_{\alpha_U} \mathbb{N}$.

Consider the algebra

$$\mathcal{I}_U := C^*(aU^m \mid a \in I_U,\, m = 0, 1, \dots).$$

Similarly, we define

$$\mathcal{I}_{U,n} := C^*(aU^m \mid a \in I_{U,n},\, m = 0, 1, \dots).$$

Then from Section 3, we know that $\mathcal{I}_U$ is an α_U and β_U invariant ideal in A_U, while $\mathcal{I}_{U,n}$ is an invariant ideal in $A_{U,n}$.

There is an alternative, equivalent definition of $\mathcal{I}_U$ which we briefly describe below even though it is not used in subsequent discussion. This alternative description does, however, illustrate some advantages when dealing with universal objects.

Let $\{E_l\}$ be the canonical basis in $\ell^2(\mathbb{Z})$. Consider the map $c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \to B\big(\ell^2(\mathbb{Z})\big)$ defined by $F \mapsto m_F$, where m_F is a bounded operator on $\ell^2(\mathbb{Z})$ given by $m_F E_l = f_\infty(l) E_l$. Additionally, let $u \in B\big(\ell^2(\mathbb{Z})\big)$ denote the standard bilateral shift $uE_l = E_{l+1}$. It is easy to check that sending $U \mapsto u$ and $M_F \mapsto m_F$ gives rise to a representation of $A_U \cong c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \rtimes_{\tilde{\alpha}_U} \mathbb{N}$ in $B\big(\ell^2(\mathbb{Z})\big)$ since u and m_F satisfy the crossed product relations. Denote this representation by π_∞. It is easily seen that $\mathcal{I}_U$ is the kernel of this representation $\mathcal{I}_U = \ker \pi_\infty$.

Next, using a convenient representation, we deduce the structure of the ideal $\mathcal{I}_U$ as described in the following statement.

Proposition 4.1. *Let $\mathcal{K}$ be the C^*-algebra of compact operators on a separable Hilbert space. We get the following isomorphisms of C^*-algebras $\mathcal{I}_U \cong C(\mathbb{Z}_s) \otimes \mathcal{K}$ and $\mathcal{I}_{U,n} \cong \mathbb{C}^{s^n} \otimes \mathcal{K}$.*

Proof. We faithfully represent A_U in $\ell^2(\mathbb{Z} \times \mathbb{Z}_{\geq 0})$, which is a more useful representation than the defining representation in $\ell^2(\mathcal{V})$. If $\{E_{k,l}\}$ is the canonical basis in $\ell^2(\mathbb{Z} \times \mathbb{Z}_{\geq 0})$, then the new representation of A_U is given by operators $\mathcal{U}$ and $\mathcal{M}_F$ for functions $F = (f_n)_{n \in \mathbb{Z}_{\geq 0}} \in c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$ given by the following formulas:

$$\mathcal{U}E_{k,l} = E_{k,l+1}, \qquad \mathcal{U}^* E_{k,l} = \begin{cases} E_{k,l-1} & \text{if } l \geq 1, \\ 0 & \text{if } l = 0, \end{cases} \qquad \text{and} \qquad \mathcal{M}_F E_{k,l} = f_l(k+l) E_{k,l}.$$

Note that $\mathcal{U}$ is the unilateral shift in the second coordinate and these operators satisfy the defining relations for the crossed product $c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \rtimes_{\tilde{\alpha}_U} \mathbb{N}$. Additionally, the following diagonal unitary operators on $\ell^2(\mathbb{Z} \times \mathbb{Z}_{\geq 0})$ given on basis elements by

$$\mathbb{U}_\theta E_{k,l} = \mathrm{e}^{2\pi \mathrm{i} \theta l} E_{k,l} \tag{4.2}$$

with $\theta \in \mathbb{R}/\mathbb{Z}$ satisfy the following relations $\mathbb{U}_\theta \mathcal{U} \mathbb{U}_\theta^{-1} = \mathrm{e}^{2\pi \mathrm{i} \theta} \mathcal{U}$ and $\mathbb{U}_\theta \mathcal{M}_F \mathbb{U}_\theta^{-1} = \mathcal{M}_F$ for every $F \in c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$. Thus $\mathbb{U}_\theta$ implements the gauge action in this representation. Since $F \mapsto \mathcal{M}_F$ is faithful, it follows from the previous section that the above representation is a faithful representation of the crossed product $A_U \cong c(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s)) \rtimes_{\tilde{\alpha}_U} \mathbb{N}$.

For $f \in C(\mathbb{Z}_s)$, let $\tilde{f}$ be a bounded sequence with values in $C(\mathbb{Z}_s)$ given by the formula $\tilde{f}_n(x) := f(x-n)$. If additionally $c \in c_0(\mathbb{Z}_{\geq 0})$, then, by the Stone-Weierstrass theorem, the linear combinations of sequences of the form $\big(\tilde{f}c\big)_n(x) = \tilde{f}_n(x) c(n)$ are dense in $c_0(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))$. Computing on basis elements we get $\mathcal{M}_{\tilde{f}c} E_{k,l} =$

$f(k)c(l)E_{k,l}$. Denoting $\mu_f E_{k,l} = f(k)E_{k,l}$ and $\nu_c E_{k,l} = c(l)E_{k,l}$, we see that $\mathcal{M}_{\tilde{f}c} = \mu_f\nu_c$. Moreover, for all $f \in C(\mathbb{Z}_s)$ and all $c \in c_0(\mathbb{Z}_{\geq 0})$ the operators μ_f commute with ν_c and with $\mathcal{U}$. Clearly, $C^*(\mu_f \mid f \in C(\mathbb{Z}_s)) \cong C(\mathbb{Z}_s)$ and, since both $\mathcal{U}$ and ν_c depend only on the second coordinate $l \in \mathbb{Z}_{\geq 0}$, we have

$$C^*(\mathcal{U}^m\nu_c \mid m \in \mathbb{Z}_{\geq 0},\, c \in c_0(\mathbb{Z}_{\geq 0})) \cong \mathcal{K}\big(\ell^2(\mathbb{Z}_{\geq 0})\big),$$

where $\mathcal{K}\big(\ell^2(\mathbb{Z}_{\geq 0})\big)$ is the algebra of compact operators on $\ell^2(\mathbb{Z}_{\geq 0})$. Finally, this implies that

$$\begin{aligned}\mathcal{I}_U &\cong C^*(\mathcal{U}^m\mathcal{M}_F \mid m \in \mathbb{Z}_{\geq 0},\, F \in c_0(\mathbb{Z}_{\geq 0}, C(\mathbb{Z}_s))\\ &= C^*(\mathcal{U}^m\nu_c\mu_f \mid m \in \mathbb{Z}_{\geq 0},\, c \in c_0(\mathbb{Z}_{\geq 0}),\, f \in C(\mathbb{Z}_s)) \cong C(\mathbb{Z}_s) \otimes \mathcal{K}\big(\ell^2(\mathbb{Z}_{\geq 0})\big).\end{aligned}$$

Restricting f's in the above argument from $C(\mathbb{Z}_s)$ to the subalgebra $C^*(M_{(n,x)})$, we get the identification $\mathcal{I}_{U,n} \cong \mathbb{C}^{s^n} \otimes \mathcal{K}\big(\ell^2(\mathbb{Z}_{\geq 0})\big)$. □

Consider the quotient $A_U/\mathcal{I}_U$ and the quotient $A_{U,n}/\mathcal{I}_{U,n}$ which, by Proposition 3.3, are isomorphic respectively to the crossed product of B_U/I_U and the crossed product of $B_{U,n}/I_{U,n}$ by the factor monomorphism $[\alpha_U]$. Since we identified B_U/I_U with $C(\mathbb{Z}_s)$ and the endomorphism $[\alpha_U]$ with $\mathfrak{a}_U$, we get the following immediate consequence of the definition of the Bunce–Deddens algebra:

Proposition 4.2. *The Bunce–Deddens algebra $BD(s^\infty)$ is isomorphic to $A_U/\mathcal{I}_U$.*

Additionally, using [18, Proposition 2.9], we get

$$A_{U,n}/\mathcal{I}_{U,n} \cong C^*(M_{(n,x)}) \rtimes_{\mathfrak{a}_u} \mathbb{Z} = BD(s^n) \cong M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big).$$

The main result of this section in the theorem below is a direct consequence of equation (3.5), Propositions 4.1 and 4.2.

Theorem 4.3. *We have the short exact sequences of C^*-algebras*

$$0 \to C(\mathbb{Z}_s) \otimes \mathcal{K} \to A_U \to BD(s^\infty) \to 0,$$

$$0 \to \mathbb{C}^{s^n} \otimes \mathcal{K} \to A_{U,n} \to M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big) \to 0. \tag{4.3}$$

4.4. ***K*-theory of A_U.** The results of the previous subsection can be used to describe the K-theory of A_U, which we do in the following proposition.

Proposition 4.4. *The K-theory of A_U satisfies the following: $K_1(A_U) \cong 0$ and*

$$0 \to C(\mathbb{Z}_s, \mathbb{Z})/\mathbb{Z} \to K_0(A_U) \to G_{s^\infty} \to 0,$$

where $G_{s^\infty} = \big\{\frac{k}{s^n} \in \mathbb{Q} \mid k \in \mathbb{Z},\, n \in \mathbb{N}\big\}$ and the subgroup of $\mathbb{Z}$ in $C(\mathbb{Z}_s, \mathbb{Z})$ is the subgroup of constant functions.

Proof. The short exact sequence (4.3) above induces the following 6-term exact sequence in K-theory:

$$\begin{array}{ccccc} K_0(\mathcal{I}_U) & \longrightarrow & K_0(A_U) & \longrightarrow & K_0(BD(s^\infty)) \\ {\scriptstyle \mathrm{ind}}\uparrow & & & & \downarrow{\scriptstyle \exp} \\ K_1(BD(s^\infty)) & \longleftarrow & K_1(A_U) & \longleftarrow & K_1(\mathcal{I}_U). \end{array}$$

Recall that $\mathcal{I}_U \cong C(\mathbb{Z}_s) \otimes \mathcal{K}$. By stability of K-theory, along with [27, Exercise 3.4], and the fact that $\mathbb{Z}_s$ is totally disconnected, we have that $K_0(\mathcal{I}_U) \cong C(\mathbb{Z}_s, \mathbb{Z})$. Since $C(\mathbb{Z}_s)$ is AF (see [10, Example III.2.5]), $K_1(\mathcal{I}_U) \cong 0$. It follows that the exponential map is zero.

By [2, Exercise 10.11.4 (b)], also see [16], $K_0(BD(s^\infty)) \cong G_{s^\infty}$ and $K_1(BD(s^\infty)) \cong \mathbb{Z}$. The generator of $K_1(BD(s^\infty))$ is a unitary and it lifts to the non-unitary isometry U in A_U. By [27, Proposition 9.2.4], the index map sends the class of the generator of $K_1(BD(s^\infty))$ to the class of $UU^* - U^*U = -P_0$. The range of the

map $\mathbb{Z} \to C(\mathbb{Z}_s, \mathbb{Z})$ is generated by $-[I - UU^*]_0$, which is seen, by direct calculation, to correspond to the constant function -1 in $C(\mathbb{Z}_s, \mathbb{Z})$. In particular, the kernel of the index map is zero. By exactness on the bottom row of the 6-term exact sequence, we see that $K_1(A_U) \cong 0$.

Since the range of the index map is the subgroup of constant functions in $C(\mathbb{Z}_s, \mathbb{Z})$, taking the quotient we can extract a short exact sequence of Abelian groups from the top row

$$0 \to C(\mathbb{Z}_s, \mathbb{Z})/\mathbb{Z} \to K_0(A_U) \to G_{s^\infty} \to 0. \qquad \square$$

The exact sequence above does not split and so does not fully determine $K_0(A_U)$. In fact, as shown below in equation (4.4), we can describe explicitly all the $K_0(A_U)$ classes but the group operation is non-obvious. To overcome this issue, we notice that A_U is the direct limit of $A_{U,n}$ and so we can, using the continuity of K-theory, write $K_0(A_U)$ as a direct limit of easier groups $K_0(A_{U,n})$, which turn out to be powers of $\mathbb{Z}$. We also calculate below the corresponding homomorphisms from $K_0(A_{U,n})$ to $K_0(A_{U,n+1})$.

First we compute the K-theory of $A_{U,n}$. This is done by the 6-term exact sequence in K-theory as above

$$\begin{array}{ccccc}
\mathbb{Z}^{s^n} \cong K_0\big(\mathbb{C}^{s^n} \otimes \mathcal{K}\big) & \longrightarrow & K_0(A_{U,n}) & \longrightarrow & K_0\big(M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big)\big) \cong \mathbb{Z} \\
\uparrow{\scriptstyle \mathrm{ind}} & & & & \downarrow{\scriptstyle \exp=0} \\
\mathbb{Z} \cong K_1\big(M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big)\big) & \longleftarrow & K_1(A_{U,n}) \cong 0 & \xleftarrow[0]{} & K_1(\mathbb{C}^{s^n} \otimes \mathcal{K}) \cong 0.
\end{array}$$

We know that $K_1\big(\mathbb{C}^{s^n} \otimes \mathcal{K}\big)$ is isomorphic to 0, thus the exponential map is 0. We know that the unitary generator of $K_1\big(M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big)\big)$ lifts to a non-unitary in $A_{U,n}$, thus the index map is injective, and exactness of the sequence forces $K_1(A_{U,n})$ to be 0. Consequently, we get a simplified short exact sequence

$$0 \to \mathbb{Z}^{s^n}/\mathbb{Z} \to K_0(A_{U,n}) \to \mathbb{Z} \to 0,$$

where the subgroup $\mathbb{Z}$ in the quotient $\mathbb{Z}^{s^n}/\mathbb{Z}$ is the diagonal subgroup. Because the right-most nontrivial group is free, the sequence necessarily splits and thus

$$K_0(A_{U,n}) \cong \big(\mathbb{Z}^{s^n}/\mathbb{Z}\big) \oplus \mathbb{Z} \cong \mathbb{Z}^{s^n}.$$

The goal is to compute the homomorphisms $\phi_n \colon K_0(A_{U,n}) \to K_0(A_{U,n+1})$ induced by inclusions $A_{U,n} \subseteq A_{U,n+1}$. First notice that $[M_{(n,x)}]$ are minimal projections in

$$A_{U,n}/\mathcal{I}_{U,n} \cong C^*(M_{(n,x)}) \rtimes_{\mathfrak{a}_u} \mathbb{Z} \cong M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big),$$

and hence their classes give the generator of $K_0\big(M_{s^n}(\mathbb{C}) \otimes C\big(S^1\big)\big) \cong \mathbb{Z}$. Specifically, in the formulas below, we choose $[M_{(n,0)}]$ for the generator in $K_0(A_{U,n})$ corresponding to that generator in $K_0(A_{U,n}/\mathcal{I}_{U,n})$.

Also, the projections $M_{(n,x)}$ generate K_0 classes of $C^*(M_{(n,x)}) \cong \mathbb{C}^{s^n}$ and, since P_0 is a minimal projection in $\mathcal{K}$, the classes $[M_{(n,x)}P_0]$ generate $K_0(\mathcal{I}_{U,n}) = K_0(C^*(M_{(n,x)}) \otimes \mathcal{K})$. Since $UU^* = I - P_0$ and $U^*U = I$, the class of P_0 in $K_0(A_{U,n})$ is zero. In the above short exact sequence, this corresponds to the fact that the diagonal subgroup of $K_0(\mathcal{I}_{U,n})$ gives a zero class in $K_0(A_{U,n})$. Thus, we choose $[M_{(n,x)}P_0]$ with $x \neq 0$ for generators of the range of the map $\mathbb{Z}^{s^n}/\mathbb{Z} \to K_0(A_{U,n})$. To summarize, we get the following explicit description of $K_0(A_{U,n})$

$$K_0(A_{U,n}) \cong \left\{ \sum_{j=1}^{s^n-1} c_j[M_{(n,j)}P_0] + k[M_{(n,0)}] \mid c_j, k \in \mathbb{Z} \right\}. \tag{4.4}$$

With those preliminaries we can now compute the maps $\phi_n\colon K_0(A_{U,n}) \to K_0(A_{U,n+1})$. Notice that the $M_{(n,x)}$'s are indicators for balls in $\mathbb{Z}_s$ so we can decompose these indicators as

$$M_{(n,x)} = \sum_{l=0}^{s-1} M_{(n+1,x+ls^n)},$$

each of which are mutually orthogonal projections. Thus an element in equation (4.4) can be written as

(4.5)
$$\sum_{j=1}^{s^n-1} c_j[M_{(n,j)}P_0] + k[M_{(n,0)}] = \sum_{j=1}^{s^n-1}\sum_{l=0}^{s-1} c_j[M_{(n+1,j+ls^n)}P_0] + k\sum_{l=0}^{s-1}[M_{(n+1,ls^n)}].$$

To compute ϕ_n, we need to write the above as combinations of generators of $K_0(A_{U,n+1})$ as in equation (4.4).

Notice that by Murray–von Neumann equivalence

$$\begin{aligned}[M_{(n+1,0)}] &= \left[U^{ls^n}M_{(n+1,0)}(U^*)^{ls^n}\right] = \left[M_{(n+1,ls^n)}U^{ls^n}(U^*)^{ls^n}\right] \\ &= [M_{(n+1,ls^n)}] - \left[M_{(n+1,ls^n)}(I - U^{ls^n}(U^*)^{ls^n})\right].\end{aligned}$$

Thus equation (4.5) may be written as

(4.6)
$$\sum_{j=1}^{s^n-1}\sum_{l=0}^{s-1} c_j[M_{(n+1,j+ls^n)}P_0] + ks[M_{(n+1,0)}] + k\sum_{l=1}^{s-1}\left[M_{(n+1,ls^n)}(I - U^{ls^n}(U^*)^{ls^n})\right].$$

Further, notice that for $l > 0$

$$I - U^{ls^n}(U^*)^{ls^n} = P_0 + P_1 + \cdots + P_{ls^n-1} = \sum_{i=0}^{ls^n-1} U^iP_0(U^*)^i.$$

By using Murray–von Neumann equivalence, the last sum in equation (4.6) is equal to

$$k\sum_{l=1}^{s-1}\sum_{i=0}^{ls^n-1}\left[M_{(n+1,ls^n)}U^iP_0(U^*)^i\right] = k\sum_{l=1}^{s-1}\sum_{i=0}^{ls^n-1}[M_{(n+1,ls^n-i)}P_0]$$

and, reindexing $ls^n - i \to i$, we get equation (4.6) as

$$\sum_{j=1}^{s^n-1}\sum_{l=0}^{s-1} c_j[M_{(n+1,j+ls^n)}P_0] + ks[M_{(n+1,0)}] + k\sum_{l=1}^{s-1}\sum_{i=1}^{ls^n}[M_{(n+1,i)}P_0].$$

Swapping the second double sum in the above equation, we get

$$\begin{aligned}\sum_{j=1}^{s^n-1}\sum_{l=0}^{s-1} c_j[M_{(n+1,j+ls^n)}P_0] + ks[M_{(n+1,0)}] + k\sum_{i=1}^{s^n-1}(s-1)[M_{(n+1,i)}P_0] \\ + k\sum_{i=s^n}^{(s-1)s^n}\left(s - \left\lceil\frac{i}{s^n}\right\rceil\right)[M_{(n+1,i)}P_0].\end{aligned}$$

Reindexing, we write this formula as

$$\sum_{i=0}^{s^{n+1}-1} c_i'[M_{(n+1,i)}P_0] + k'[M_{(n+1,0)}],$$

where $k' = ks$ and

$$c'_i = \begin{cases} c_{(i \bmod s^n)} + k(s-1), & \text{if } 1 \leq i < s^n, \\ c_{(i \bmod s^n)} + k\left(s - \left\lceil \dfrac{i}{s^n} \right\rceil\right), & \text{if } s^n \leq i \leq (s-1)s^n, \\ c_{(i \bmod s^n)}, & \text{if } (s-1)s^n < i < s^{n+1}. \end{cases} \tag{4.7}$$

Thus the map $\phi_n \colon K_0(A_{U,n}) \to K_0(A_{U,n+1})$, using the identification $K_0(A_{U,n}) \cong \mathbb{Z}^{s^n}$ from equation (4.4), is given by

$$\phi_n((c_1, \ldots, c_{s^n-1}, k)) = (c'_1, \ldots, c'_{s^{n+1}-1}, k').$$

As a consequence, we get the following proposition.

Proposition 4.5. *The K-theory of A_U is given by $K_0(A_U) = \varinjlim \mathbb{Z}^{s^n}$ with maps $\phi_n \colon \mathbb{Z}^{s^n} \to \mathbb{Z}^{s^{n+1}}$ given by*

$$\phi_n((c_1, \ldots, c_{s^n-1}, k)) = (c'_1, \ldots, c'_{s^{n+1}-1}, ks),$$

where c'_i are described in equation (4.7).

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